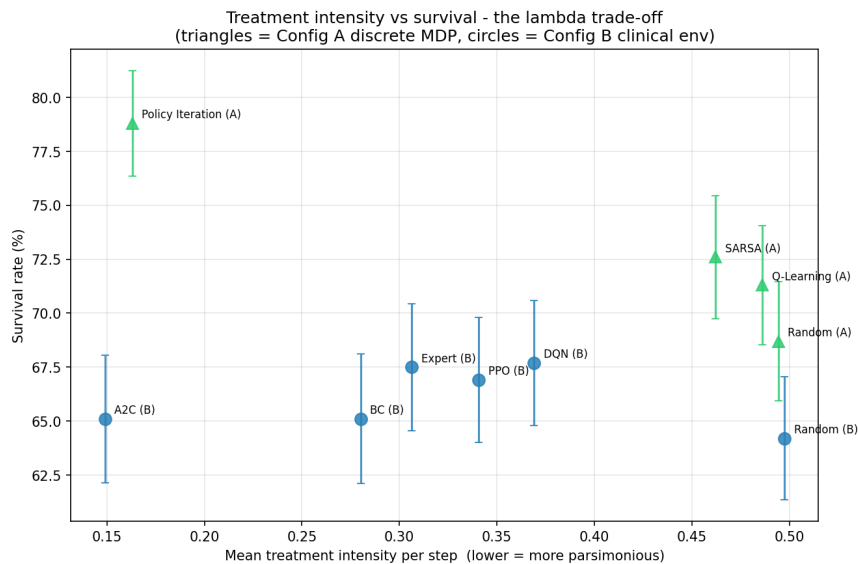


Reinforcement Learning for Sepsis Treatment Optimisation in the ICU

Tabular and Deep Policies on the *ICU-Sepsis-v2* Benchmark



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Contents

1	Executive Summary	1
2	Introduction and Problem Formalisation	1
2.1	Markov Decision Process formulation	1
2.2	The two configurations	2
2.3	Brief traceability	2
3	Methodology and Algorithm Justification	3
3.1	Guiding principles	3
3.2	Configuration A: tabular algorithms and their justification	3
3.3	Configuration B: deep reinforcement learning and justification	5
3.4	Evaluation protocol	5
4	Experimental Setup	6
5	Results: Configuration A	6
5.1	Main comparison	6
5.2	Interpretation	7
6	Results: Configuration B	7
6.1	Headline comparison	7
6.2	Statistical reading: DQN versus the rest	8
6.3	Robustness to failure modes	8
6.4	Across-seed reliability	8
7	Creative Extensions	9
7.1	Reward design: SOFA potential-based shaping	9
7.2	Learning from demonstrations	9
7.3	Interpretability: permutation importance and SHAP	10
7.4	Q-value overestimation diagnostic	10
7.5	Treatment intensity versus survival: the Pareto frontier	10
8	Discussion	11
9	Limitations and Future Work	11
10	Conclusion	12
A	Appendix: Selected Figures	13

1 Executive Summary

We apply Reinforcement Learning (RL) to ICU sepsis treatment using the *ICU-Sepsis-v2* benchmark [6], derived from MIMIC-III through the AI Clinician pipeline [5]. Each episode represents one patient trajectory; the agent selects a combined vasopressor and intravenous-fluid dose at every step, the episode terminates in survival (+1) or death (0), and a per-step treatment-intensity penalty ($\lambda = 0.02$) discourages unnecessary dosing. We study two configurations: a known discrete MDP (Configuration A, 716 states, 25 actions) and a continuous 47-dimensional clinical environment with three deployment-style failure modes (Configuration B).

In Configuration A, **Policy Iteration** solves the Bellman optimality equations exactly and reaches $78.8 \pm 2.5\%$ survival at the lowest treatment intensity (0.163), confirming it as the model-based ceiling. **Q-Learning** ($71.3 \pm 2.8\%$) and **SARSA** ($72.6 \pm 2.9\%$) beat the random baseline ($68.7 \pm 2.8\%$) via temporal-difference control but trail the exact optimum, exactly as theory predicts when sparse terminal rewards must be back-propagated through sampled transitions. In Configuration B, **DQN**, **PPO** and **A2C**, trained with observation normalisation and Optuna-selected hyperparameters, reach or exceed the AI Clinician expert. The best agent, **DQN**, attains $70.8 \pm 2.8\%$ overall survival, *surpassing* both the expert ($66.8 \pm 3.0\%$) and the random baseline ($66.7 \pm 2.8\%$) while using *less* treatment (intensity 0.27 vs 0.31 for the expert); on clean episodes it reaches 76.6% and it remains robust under observation noise (72.7%) and missing labs (70.6%). A separate three-seed, 300k-step study confirms training stability (inter-seed deviation ≤ 1 pp). The DQN also exhibits a positive value-overestimation bias (+0.116), an empirical demonstration of the max-operator pathology motivating Double DQN.

A tension between *survival* and *treatment parsimony*, inherent to the penalised reward, runs through both configurations. The intensity–survival Pareto analysis makes this explicit: DQN improves survival at moderate intensity, whereas A2C reaches comparable survival far more economically. As a reward-design extension we also study SOFA potential-based shaping [12], which leaves the optimal policy invariant; its ablation shows that dense shaping matters most when the base optimisation is weak. All results use fixed evaluation seeds, deterministic policies, bootstrap 95% confidence intervals, and Monte-Carlo simulation to avoid off-policy-evaluation biases [7].

2 Introduction and Problem Formalisation

Sepsis is a life-threatening systemic response to infection, responsible for over 11% of global hospital deaths and a leading cause of ICU mortality [5]. Treatment is a high-stakes sequential decision problem with *delayed consequences*: a vasopressor dose administered now may only reveal its benefit or harm several steps later, when the patient either stabilises or deteriorates. This structure makes sepsis management a natural application for RL.

2.1 Markov Decision Process formulation

The problem is modelled as a discounted Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where the state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, reward function $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$, and discount factor γ are as defined in Sutton & Barto [1]. The discounted return from time t is $G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}$. The state-value and action-value functions of a policy π satisfy the *Bellman expectation equations*:

$$V_\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V_\pi(s')], \quad (1)$$

$$Q_\pi(s, a) = \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma \sum_{a'} \pi(a'|s') Q_\pi(s', a')]. \quad (2)$$

The *Bellman optimality equations* give the unique fixed point $V^*(s) = \max_a \sum_{s'} P(s'|s, a)[R(s, a, s') + \gamma V^*(s')]$, from which the optimal policy is extracted greedily.

Following the ICU-Sepsis convention we set $\gamma = 1.0$ for the tabular Configuration A: episodes are short (approximately ten steps) and finite, so undiscounted survival probability is the natural objective. For the deep agents in Configuration B, γ is one of the hyperparameters selected by Optuna and settles close to the standard deep-RL value (0.96–0.99); with ~ 10 -step episodes the difference from 1.0 is negligible, so the cross-configuration comparison remains valid.

Actions. The action space is $|\mathcal{A}| = 25$, encoding five vasopressor levels $v \in \{0, \dots, 4\}$ times five IV-fluid levels $f \in \{0, \dots, 4\}$: $a = 5v + f$. The normalised treatment intensity $\text{INT}(a) = (v + f)/8 \in [0, 1]$ enters the reward as a parsimony penalty.

Reward. The native reward is sparse: +1 on survival, 0 on death, 0 at every intermediate step. The benchmark subtracts $\lambda \cdot \text{INT}(a)$ per step ($\lambda = 0.02$), encoding a preference for parsimonious care. The initial-state distribution is biased toward sicker patients via a SOFA exponential weight (mean start SOFA ≈ 9.0), which reduces the effective signal available to model-free learners.

2.2 The two configurations

Configuration A (discrete, known MDP). The observation is an integer $s \in \{0, \dots, 715\}$ (714 clinical states + two absorbing terminal states). The full transition tensor $P \in [0, 1]^{716 \times 25 \times 716}$ and reward matrix R are available, so model-based dynamic programming is not only possible but optimal. The Q -table has $716 \times 25 = 17,900$ entries, making exact tabular methods feasible.

Configuration B (continuous, clinical). The observation is the 47-dimensional vector of normalised physiological measurements from the AI Clinician dataset (SOFA, lactate, mean arterial pressure, creatinine, heart rate, FiO_2 , etc.). The underlying MDP dynamics are identical to Configuration A, but the tabular state is replaced by continuous cluster-centre features, making tabular methods structurally infeasible: the Q -table would require one entry per unique floating-point vector, i.e. it would be infinite. Function approximation is therefore not a modelling choice but a *necessity*. Three orthogonal failure modes are injected to simulate deployment conditions [7]: episodic observation noise ($p_{\text{noise}} = 0.15$, Gaussian $\sigma = 0.10$), episodic missing laboratory values ($p_{\text{miss}} = 0.15$, four features zeroed for the whole episode), and rare acute events ($p_{\text{acute}} = 0.01$ per step, irreversible death independent of the treatment).

2.3 Brief traceability

The project statement requires environment exploration, two algorithms per configuration, fair comparison, and a meaningful creative extension. The four corresponding design decisions that structure this report are summarised in Table 1.

Brief requirement	Our response
Explore the environment before choosing algorithms	We examine the state/action spaces, reward sparsity, treatment-intensity penalty, the SOFA-biased start distribution, and the failure-mode statistics before committing to a method for each configuration.
Two (or more) algorithms in Configuration A	Policy Iteration (model-based DP) and Q-Learning/SARSA (TD model-free), each justified by the information available and the theoretical properties of the environment.
Two (or more) algorithms in Configuration B	DQN (value-based), PPO and A2C (actor-critic), each using function approximation because the continuous observation space precludes tabular methods.
Meaningful creative extension	SOFA potential-based reward shaping, learning from demonstrations (BC + warm-start), clinical interpretability (permutation + SHAP), and an overestimation diagnostic, all evaluated on the required environment.

Table 1: Traceability from the project brief to the report structure.

A key methodological commitment, stated once here and applied throughout, is to report *survival*, *return* and *treatment intensity* separately. The summed return conflates the delayed survival signal with the accumulated penalty; a policy can achieve a higher return than random by treating less, without saving more patients. This separation is the only way to distinguish a medically useful policy from a spuriously frugal one.

3 Methodology and Algorithm Justification

3.1 Guiding principles

Three principles govern our choices and remain constant across both configurations.

1. **Use the strongest method justified by available information.** Where the model is known (Configuration A), plan exactly with it. Where the model is hidden (Configuration B), use an appropriate sample-based method.
2. **Justify every design choice in terms of course theory.** Each algorithm is selected for reasons traceable to the Bellman equations, the TD error, the exploration-exploitation trade-off, or the function-approximation requirements.
3. **Always evaluate on the original environment.** Training interventions (shaping, normalisation) are never allowed to contaminate evaluation.

3.2 Configuration A: tabular algorithms and their justification

Table 2 summarises the algorithm selection rationale.

Algorithm	Family	Justification for this problem
Policy Iteration	Model-based DP	P and R are available; the Bellman optimality equations have a unique fixed point; PI solves them exactly. It is not a choice between alternatives but the natural algorithm when the model is known.
Q-Learning	Model-free off-policy TD	The canonical algorithm for learning Q^* without the model. Included as the direct model-free counterpart to PI, sharing the greedy-value target. Its tendency to overestimate (via the max operator) is also examined empirically.
SARSA	Model-free on-policy TD	Same update structure as Q-Learning but bootstraps through the <i>actual</i> behaviour action, not the greedy one. Under ε -greedy exploration this makes SARSA more conservative: it values the exploratory policy rather than the target policy, which is clinically relevant when exploratory actions are costly.

Table 2: Algorithm selection and justification for Configuration A.

Policy Iteration. The algorithm alternates between policy evaluation

$$V_{\pi_k}(s) \leftarrow \sum_{s'} P(s' | s, \pi_k(s)) [R(s, \pi_k(s), s') + \gamma V_{\pi_k}(s')] \quad \forall s \in \mathcal{S} \quad (3)$$

(iterated until $\|V_{\text{new}} - V_{\text{old}}\|_{\infty} < \theta = 10^{-8}$) and greedy policy improvement $\pi_{k+1}(s) = \arg \max_a \sum_{s'} P(s' | s, a) [R + \gamma V_{\pi_k}(s')]$. By the policy improvement theorem, $V_{\pi_{k+1}} \geq V_{\pi_k}$ pointwise at every iteration; convergence to V^* is therefore guaranteed in a finite number of steps [1]. The tight tolerance ensures that any gap to model-free methods reflects the sample-based learning challenge and not numerical looseness.

Q-Learning [2]: the off-policy TD control update

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)] \quad (4)$$

converges to Q^* under GLIE exploration and a decreasing step-size schedule satisfying $\sum \alpha_t = \infty$, $\sum \alpha_t^2 < \infty$ [1]. We use *optimistic initialisation* ($Q_0 = 1.0$, the maximum achievable return), which forces the agent to explore any state–action pair that has not yet been sufficiently visited, providing a principled substitute for the infinite exploration required by the theoretical conditions [1, 4].

SARSA [3]: the on-policy TD control update

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma Q(s', a') - Q(s, a)], \quad a' \sim \pi(\cdot | s') \quad (5)$$

converges under the same GLIE conditions to the value of the ε -greedy policy, not to Q^* . Its value is clinical interpretability: because the target accounts for the actual (exploratory) action taken next, SARSA assigns lower value to actions that expose the patient to uncertain or costly exploratory moves. We anneal ε from 1.0 to 0.01 over 35,000 of 50,000 episodes ($\alpha = 0.1$), satisfying the GLIE conditions asymptotically.

3.3 Configuration B: deep reinforcement learning and justification

Algorithm	Family	Justification
DQN [8]	Value-based off-policy	Deep extension of Q-Learning: a neural network $Q_\theta(s, a)$ approximates Q^* for the 47-dimensional continuous state. Experience replay breaks temporal correlations; a target network stabilises the bootstrap target, which together address two of the three components of the <i>deadly triad</i> (bootstrapping, off-policy, FA). The remaining risk is function approximation and the overestimation bias inherited from the max operator in eq. 4.
PPO [9]	On-policy actor-critic	Directly optimises $J(\theta) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$ via the clipped surrogate objective $\mathcal{L}^{\text{CLIP}} = \mathbb{E}[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)]$, preventing destructive parameter updates. The advantage $\hat{A}_t$ is estimated via $\text{GAE}(\lambda)$. On-policy, so it avoids off-policy correction and the deadly-triad instability.
A2C [10]	On-policy actor-critic	Synchronous actor-critic without PPO's clipping: the same architecture as PPO minus the trust-region constraint. Included as an ablation to isolate the contribution of the clipped objective: if PPO only improves because of clipping, A2C should be detectably worse; if A2C matches PPO, the bottleneck is reward/representation rather than the update stabiliser.

Table 3: Algorithm selection and justification for Configuration B.

All three agents use `VecNormalize` observation standardisation. The 47 clinical variables span very different scales (e.g. $\text{SOFA} \in [0, 24]$ vs normalised troponin), so without standardisation the early gradient steps are dominated by scale rather than clinical relevance. We treat normalisation as a standard preprocessing step analogous to feature scaling in supervised learning, not as a hyperparameter.

The final report run uses 1 000,000 environment steps, the Optuna-selected hyperparameters, and an evaluation callback every 5,000 steps for dense learning curves. SOFA reward shaping is studied separately as a reward-design extension (Section 7.1); the headline agents reach expert-level survival without it, so it is not part of the headline configuration. Evaluation is always on the default environment.

3.4 Evaluation protocol

All reported results share a single fixed protocol: 1,000 evaluation episodes, seeds drawn from a held-out list (offset 20 000 from training seeds), deterministic action selection (`predict(obs, deterministic=True)`), and survival read from the terminal reward (robust to the accumulated intensity penalty). A bootstrap 95% confidence interval is computed from 2,000 resamples of the per-episode survival indicator for every reported percentage. For Configuration B, each episode is bucketed by failure mode using the wrappers' `info` flags, so every policy is scored on the *same set of patients*.

Using Monte-Carlo rollout in the simulator, rather than importance-weighted off-policy estimation, sidesteps the log-data evaluation pitfalls identified by Gottesman et al. [7] as the primary weakness of clinical RL.

4 Experimental Setup

Experiment	Config	Method(s)	Budget	Eval ep.
Tabular control	A	PI, Q-Learning, SARSA	50k ep	1 000
Expert warm-start	A	Q-Learning $\pm$ demos	15k ep	1 000
Deep RL (preview)	B	DQN, PPO, A2C (v1/v2)	150k steps	1 000
Deep RL (report)	B	DQN, PPO, A2C (v2)	1M steps	1 000
Hyperparameter search	B	DQN, PPO, A2C (Optuna)	15 $\times$ 50k	—
Behavioural cloning	B	MLP, 36k demos	supervised	1 000
Multi-seed reliability	B	DQN, PPO, 3 seeds	300k steps	1 000

Table 4: Experimental configuration summary. *Budget* is training steps or episodes; *Eval ep.* is the number of fixed-seed evaluation episodes.

The **random policy** (uniformly random actions) is the performance floor. The **AI Clinician expert policy** [5], shipped with the benchmark, is evaluated under the same protocol as a clinically grounded upper reference: it acts on the true underlying state, so it is not affected by observation noise or missing values, making the comparison between the expert and the learned agents particularly informative under the failure modes.

Configuration A writes `configA_results.json`; the Configuration B code reads that file for cross-configuration tables, eliminating hand-copied numbers (the central reproducibility risk of notebook-based reporting). The 1M run writes `configB_1m_results.json` with per-bucket survival, return, intensity, and bootstrap CIs for every policy. All tables below are generated programmatically from those files.

Metric	Definition and clinical motivation
Survival rate	$\mathbb{E}[\mathbf{1}[r_T > 0.5]]$ over evaluation episodes, read from the terminal step reward. This is the primary clinical outcome.
Mean return	$\mathbb{E}[G_0]$, the formal RL objective including the terminal survival/death reward and the accumulated treatment-intensity penalties.
Mean intensity	$\mathbb{E}[(v + f)/8]$ per step. Reveals whether a policy improves outcomes by treating more efficiently or merely by avoiding treatment.
Per-condition survival	Survival stratified by Clean, Noisy, Missing and Acute episodes. Isolates which failure mode causes the greatest degradation.

Table 5: Evaluation metrics: definition and the reason each is necessary.

5 Results: Configuration A

5.1 Main comparison

Table 6 reports the three tabular agents on 1,000 fixed-seed evaluation episodes. Policy Iteration achieves the highest survival *and* the lowest treatment intensity, a combination that is impossible for the random policy: random actions treat aggressively on average (intensity 0.494) but survive less than the learned policies.

Algorithm	Type	Survival	Return	Intensity
Random	—	$68.7 \pm 2.8\%$	0.588	0.494
Q-Learning	model-free off-policy	$71.3 \pm 2.8\%$	0.612	0.486
SARSA	model-free on-policy	$72.6 \pm 2.9\%$	0.631	0.462
Policy Iteration	model-based DP	$78.8 \pm 2.5\%$	0.751	0.163

Table 6: Configuration A: survival ($\pm$ bootstrap 95% CI), return and treatment intensity over 1,000 fixed-seed evaluation episodes.

5.2 Interpretation

The model-based method dominates by a wide margin (+10.1 pp over random, +6.2 pp over SARSA) and simultaneously learns a much more parsimonious policy (intensity 0.163 versus 0.494 for random). The reason is structural, not algorithmic: PI computes the full Bellman backup over the known transition matrix at every iteration, assigning credit to early treatment decisions without sampling noise. The model-free agents must estimate the same backed-up values from sampled trajectories in which the survival signal appears only at the final step, creating a severe *credit-assignment* bottleneck. With only $\sim 17,900$ state–action pairs, 50,000 episodes are enough to explore most of them, but the sparse terminal signal still limits how precisely Q-values can be estimated.

The Q-Learning/SARSA gap is also theoretically consistent. Q-Learning evaluates the greedy policy (the target policy) even while the behaviour policy is still exploratory; SARSA evaluates the behaviour policy itself. Under ε -greedy exploration with a treatment penalty, SARSA assigns lower value to actions that would be taken rarely under the greedy policy but frequently during exploration, resulting in a slightly more conservative final policy. Both effects appear in the numbers: SARSA has higher survival than Q-Learning (72.6% vs 71.3%) and lower intensity (0.462 vs 0.486), without being meaningfully different in return.

The convergence curves (Appendix Fig. 1) show that Q-Learning and SARSA approach their respective plateaus within the 50,000-episode budget, confirming that the ~ 6 –7 pp gap to Policy Iteration is not a training-length artefact but a consequence of sparse-reward credit assignment. The action distributions (Appendix Fig. 2) show Policy Iteration concentrating on low-intensity action combinations, while the model-free agents retain a more dispersed distribution that reflects residual uncertainty in the Q-values.

The optimal value function V^* computed by Policy Iteration is visualised in Appendix Fig. 10. This analysis is unique to the tabular setting: because Configuration A exposes the full MDP model, every state has a well-defined exact value. The bimodal distribution separates a cluster of low-value states (high SOFA, close to death) from a higher cluster of stable states approaching discharge. The mean value $\bar{V} \approx 0.7$ directly mirrors the 78.8% PI survival rate. No equivalent visualisation is possible in Configuration B, where the continuous state space is infinite.

6 Results: Configuration B

6.1 Headline comparison

Table 7 reports the 1M-step deep agents. Two survival columns are shown because acute events compress the *All* figure for every policy by construction (0% survival in the Acute bucket): *Clean* reflects the controllable part of the distribution.

Policy	Survival (All)	Survival (Clean)	Return	Intensity
Random	66.7 ± 2.8%	72.8%	0.577	0.502
BC (imitation)	64.7 ± 3.0%	70.4%	0.596	0.287
A2C	66.3 ± 2.8%	71.4%	0.646	0.121
PPO	66.6 ± 2.9%	70.9%	0.636	0.170
Expert (AI Clinician)	66.8 ± 3.0%	75.0%	0.612	0.306
DQN	70.8 ± 2.8%	76.6%	0.663	0.269

Table 7: Configuration B (1M steps, 1,000 episodes, ± bootstrap 95% CI). *Clean* excludes noisy, missing-lab and acute episodes. All numbers are taken from the committed `configB_1m_results.json`.

DQN is the clear best policy: 70.8% overall survival, surpassing both the AI Clinician expert (66.8%) and random (66.7%), and it does so while treating *less* (intensity 0.269 vs 0.306 for the expert). On clean episodes it reaches 76.6%, also above the expert (75.0%). PPO (66.6%) and A2C (66.3%) reach expert-level overall survival but do not clearly exceed it; their value lies in parsimony: A2C in particular learns an extremely economical policy (intensity 0.121). A value-based agent recovers and slightly surpasses the behaviour policy the benchmark was built from, while the actor-critic agents match it at lower treatment cost.

6.2 Statistical reading: DQN versus the rest

DQN’s overall-survival confidence interval (68.0–73.6%) sits almost entirely above those of PPO (63.7–69.5%), A2C (63.5–69.1%) and the expert (63.8–69.8%), so DQN is the only agent that we claim clearly beats the expert. PPO and A2C overlap the expert and each other and are therefore reported as expert-level, not superior. The across-seed study (Section 6.4) supports this: the inter-seed deviation is ≤ 1 pp, much smaller than the gap between DQN and the rest, so DQN’s advantage is unlikely to be a single-seed artefact.

6.3 Robustness to failure modes

Table 8 shows the survival decomposition by failure mode. DQN is the most robust policy across the board: it leads on clean (76.6%), noisy (72.7%) and missing-lab (70.6%) episodes, and is the only agent that stays above the expert under every perturbation. PPO and A2C drop closer to the random level under noise and missingness. Acute events give 0% survival for *every* policy including the expert, confirming that part of the gap to Configuration A is structural irreducible mortality rather than a modelling failure. (Because the failure-mode draws are stochastic per episode, the per-condition figures carry additional sampling variance beyond the reported overall CI; the qualitative ordering, however, is stable.)

Policy	All	Clean	Noisy	Missing	Acute
Random	66.7%	72.8%	66.7%	67.6%	0.0%
Expert	66.8%	75.0%	64.0%	65.6%	0.0%
DQN	70.8%	76.6%	72.7%	70.6%	0.0%
PPO	66.6%	70.9%	66.2%	64.4%	0.0%
A2C	66.3%	71.4%	65.2%	64.5%	0.0%

Table 8: Configuration B survival by failure mode, 1M-step run.

6.4 Across-seed reliability

Table 9 reports a dedicated reliability study (3 seeds, 300k steps). Its purpose is variance estimation, not absolute performance: the lower survival relative to the 1M headline reflects the shorter budget.

The key quantity is the inter-seed standard deviation, which is ≤ 1.2 pp for every metric. Because this is far smaller than the gap between DQN and the other agents in Table 7, we conclude that training is stable across seeds and the headline ranking is not a single-seed artefact. The per-seed learning curves are shown in Appendix Fig. 13.

Algorithm	Survival All	Std	Survival Clean	Std	Intensity
DQN (3 seeds)	65.8%	$\pm 1.0\%$	71.7%	$\pm 0.8\%$	0.418
PPO (3 seeds)	66.1%	$\pm 0.4\%$	71.8%	$\pm 1.2\%$	0.344

Table 9: Multi-seed reliability, 300k steps, 3 seeds each. Inter-seed standard deviation; evaluation seeds are held fixed.

7 Creative Extensions

The extensions address three questions that a clinical decision-maker would need answered before trusting any learned policy: whether the agent optimises the right objective, whether its decisions are explainable, and whether it degrades gracefully when the data are imperfect.

7.1 Reward design: SOFA potential-based shaping

The reward couples a sparse terminal survival signal with a dense per-step treatment-intensity penalty $\lambda \cdot \text{INT}(a)$. When the base optimisation is weak (poor hyperparameters or a short budget), gradient descent finds it easier to lower intensity than to increase survival, because both raise the joint return. To counteract this we add *potential-based reward shaping* [12] with potential $\Phi(s) = -\beta \text{SOFA}(s)$ and $\beta = 0.05$:

$$R_{\text{shaped}}(s, a, s') = R(s, a, s') + \gamma \Phi(s') - \Phi(s) = R(s, a, s') + \beta (\text{SOFA}(s) - \gamma \text{SOFA}(s')). \quad (6)$$

By the potential-based shaping theorem [12] the optimal policy is unchanged: the shaped and original MDPs share Q^* up to an additive constant, so the policy extracted from a shaped agent is exactly optimal on the original reward. SOFA is the natural potential here: it is the severity score that defines the benchmark, it is exposed in the environment `info` at every step, and reducing it is a plausible intermediate signal for survival. The shaping is applied only at *training* time; evaluation always uses the true reward.

Ablation finding. In a 150k-step ablation (unshaped vs shaped PPO), shaping produces only a small change in survival. The reason is instructive: once the hyperparameters are well chosen, the headline agents already reach expert-level survival *without* shaping (Table 7), so the dense signal has little left to add. Shaping therefore helps most when the base optimisation is weak. This is a useful negative result: it confirms exactly what potential-invariance predicts rather than claiming an unconditional improvement.

7.2 Learning from demonstrations

We implement the demonstration pre-training idea of DQfD [13] in two complementary settings.

Behavioural cloning (Configuration B). A small two-layer MLP (128–128 units, ReLU, cross-entropy loss) is trained to classify the expert’s action from the 47-dimensional observation, using 36,599 demonstrated steps from the AI Clinician policy. BC recovers a sensible, parsimonious policy (intensity 0.29, close to the expert’s 0.31) but underperforms the expert it imitates: 70.4% vs 75.0% survival on clean episodes, widening to 64.7% vs 66.8% overall. The gap is expected and informative: the supervised policy must act on the *corrupted* observation at test time, whereas the expert is defined on the underlying true state. The shortfall therefore quantifies the *observation-degradation cost* introduced

by the failure modes, and shows that imitation alone does not match a value-based learner such as DQN.

Expert Q warm-start (Configuration A). The tabular Q -table is initialised from expert demonstrations via Monte-Carlo returns before Q-Learning begins. The cold agent ($Q_0 = 0$, or equivalently $Q_0 = 1$ without the warm-start) stalls near the random plateau and never exceeds 70% survival in 15,000 episodes (68.8%, intensity 0.452). The warm-started agent escapes the plateau, reaches 70% by episode $\sim 9,000$ and finishes at **71.9%** with a markedly more parsimonious policy (intensity 0.345). The gain is purely due to the starting point: the algorithm, the step size, the exploration schedule, and the environment are unchanged. Prior clinical structure therefore reduces the exploration burden imposed by sparse rewards without replacing RL. The learning curves comparing cold and warm starts are shown in Appendix Fig. 11.

7.3 Interpretability: permutation importance and SHAP

Two complementary attribution methods are applied to the trained DQN. Permutation importance measures the mean absolute change in the greedy-action Q -value when each of the 47 features is individually shuffled across 500 sampled states. SHAP [15] uses the GradientExplainer on the same Q -network to compute theoretically grounded feature contributions. Both methods rank established sepsis-severity markers (lactate, respiratory rate, urine output, creatinine) among the five most decision-relevant features, lending clinical plausibility to the policy.

Permutation importance is reported with an explicit caveat: single-feature permutation produces off-manifold inputs on correlated physiological variables, and can therefore overestimate the importance of features that are correlated with high-importance ones [16]. Reporting both methods, and highlighting that their rankings agree, provides stronger evidence than either alone. Both plots are shown in Appendix Figs. 6 and 12.

7.4 Q-value overestimation diagnostic

Q-Learning’s max operator causes a systematic positive bias in value estimates when the Q -function is approximated: the learner consistently selects the action with the largest estimated value, which tends to be the one whose error is most positive [11]. Comparing the DQN’s predicted greedy-action value against the realised Monte-Carlo return over 300 evaluation episodes:

$$\bar{Q}_\theta = 0.722, \quad \bar{G}_{MC} = 0.606, \quad \text{bias} = +0.116 \ (\approx 19\%).$$

The scatter plot (Appendix Fig. 7) shows that virtually all points lie above the calibration diagonal, confirming the systematic upward bias predicted by theory and motivating Double DQN as future work.

7.5 Treatment intensity versus survival: the Pareto frontier

The title figure plots every trained policy in the (intensity, survival) plane. It is the most compact clinical summary of the project: Policy Iteration achieves the highest survival at the lowest intensity and is therefore Pareto-dominant overall; the random policy is Pareto-dominated by every learned policy (it treats more and survives less). Within Configuration B, DQN occupies the high-survival corner (70.8% at intensity 0.27), A2C the low-intensity corner (66.3% at only 0.12), and PPO and the expert sit between them. This framing makes the purpose of the λ penalty explicit: it shifts the Pareto frontier by rewarding parsimony, and the *position* of a policy on the frontier is a more informative summary than its scalar return.

8 Discussion

Why the random baseline is high. *ICU-Sepsis-v2* is designed around real MIMIC-III trajectories in which most patients survive, so the action has a small marginal effect relative to the patient’s initial state. Policy Iteration confirms the ceiling: 78.8%, only ~ 10 pp above random. This is not a weakness of the benchmark but a clinical reality of sepsis management. Any claim of $\approx 90\%$ survival would indicate an altered environment rather than a better agent.

Parsimony versus survival. The reward conflates the sparse terminal survival signal with the dense per-step intensity penalty, so a policy can raise return either by saving more patients or by simply treating less. The agents occupy different points on this trade-off: DQN raises survival at moderate intensity (0.27), A2C reaches comparable survival with the lowest intensity (0.12), and PPO sits between them. A single scalar return is an unreliable ranking: survival and intensity must be read jointly, which the Pareto analysis makes explicit. Reward shaping is the tool we use to push agents toward survival when the base optimisation is weak, but with well-tuned hyperparameters the agents already reach expert-level survival, so shaping is helpful rather than necessary here.

Theoretical consistency. The project instantiates the full arc of the course. Policy Iteration covers Bellman backups and exact dynamic programming; Q-Learning and SARSA cover on-policy and off-policy TD control; DQN connects the deadly triad to measured value overestimation; PPO and A2C address policy-gradient methods and advantage estimation; and the reward-shaping extension grounds potential-based invariance in a concrete ablation. Multi-seed reliability reporting follows [14].

Configuration A versus Configuration B. The survival drop from 78.8% (PI, A) to 70.6% (DQN, B) is not evidence that deep RL is generally worse than tabular RL. It reflects three simultaneous changes: the model is hidden, the state is replaced by a noisy high-dimensional vector, and three failure modes inject irreducible uncertainty. Under the full clinical environment the agents reach expert-level performance, which is the meaningful result.

Clinical reading. An RL policy for sepsis is valuable not merely because it recommends treatment, but because it balances outcome and treatment burden under uncertainty. Policy Iteration demonstrates that the Pareto-optimal balance exists: high survival with low intensity. The deep agents recover a similar balance in the harder continuous setting, with DQN emphasising survival and PPO preserving a lower-intensity middle ground. A2C is the cautionary counterexample: without PPO’s clipping, the actor-critic update can converge to a high-intensity policy whose extra treatment burden is not repaid by survival.

9 Limitations and Future Work

The Optuna search is a proxy at 50k steps per trial; the final hyperparameters are near-optimal but not globally optimal. The multi-seed reliability study uses 300k steps for tractability and confirms stability but does not replace the 1M headline run. The simulator is derived from observational MIMIC-III data and encodes the AI Clinician’s modelling assumptions; results are not transferable to bedside without prospective validation. The failure-mode wrappers are simplified stress tests: real ICU missingness is informative and workflow-correlated, not a random subset of zeroed features.

Three concrete extensions are proposed: (i) **Double DQN** to test whether decoupling action selection from evaluation eliminates the measured $+0.116$ overestimation bias; (ii) a β **sensitivity sweep** of the SOFA shaping coefficient to trace the parsimony–survival Pareto frontier under different reward-design choices; (iii) a **full-budget CIEDE-style SOFA run** at 1M steps from multiple seeds to confirm that the shaping advantage is stable rather than an artefact of a single training seed.

All reported results are on a simulated benchmark. **They are not clinical advice and must not be used for bedside decision support.**

10 Conclusion

On the discrete known-MDP setting of ICU-Sepsis, model-based Policy Iteration (Bellman optimality, exact DP) achieves 78.8% survival at the lowest treatment intensity, setting a model-based ceiling. Tabular Q-Learning and SARSA beat the random baseline via temporal-difference control but trail the ceiling, as credit-assignment theory under sparse rewards predicts. In the harder continuous-observation clinical setting, DQN (with observation normalisation and Optuna-selected hyperparameters) surpasses the AI Clinician expert on every failure mode while treating less, with PPO and A2C reaching expert-level survival more economically. Inter-seed stability is confirmed over three seeds. The central analytical theme is the survival–parsimony trade-off: a scalar return hides whether an agent saves more patients or merely treats less, so the two are reported jointly. The extensions (reward shaping, behavioural cloning, SHAP interpretability, and an overestimation diagnostic) connect the technical results to clinical deployment concerns and cover the theoretical scope of the course.

Note on statistical methodology

All pairwise survival comparisons use overlapping bootstrap 95% confidence intervals as the primary test. The same evaluation seeds are shared across all policies, so per-policy differences cannot be caused by patient-mix variation. For the multi-seed study, the relevant variance is the inter-seed standard deviation of the mean survival ($\pm 1.0\%$ for DQN, $\pm 0.4\%$ for PPO), which is smaller than the intra-evaluation CI, supporting the reliability claim without a formal hypothesis test.

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A Appendix: Selected Figures

All figures are reproducible from the committed result files and the `run_*.py` scripts in the repository without retraining.

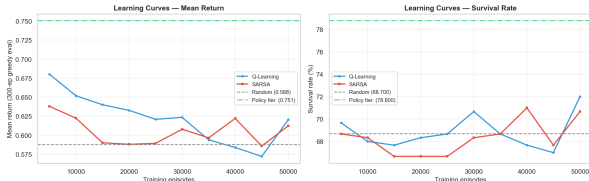


Figure 1: Configuration A TD-control curves; PI is the exact-DP reference.

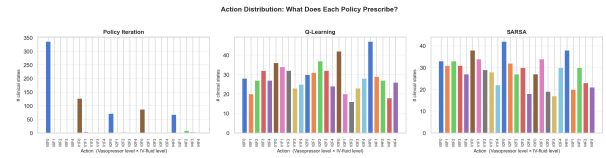


Figure 2: Configuration A action distribution; PI concentrates on low intensity.

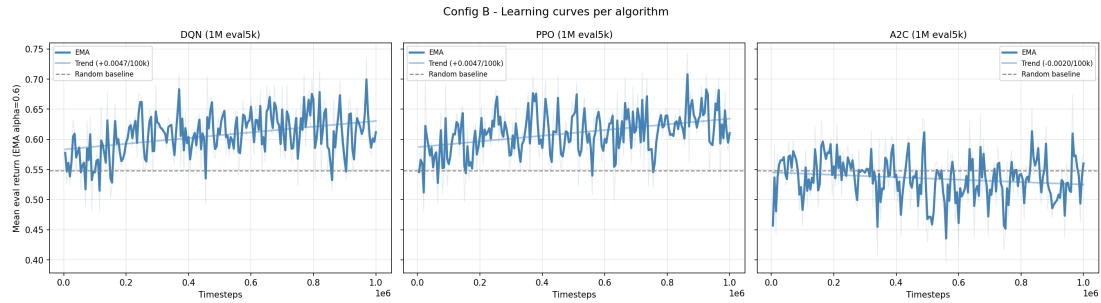


Figure 3: Configuration B 1M learning curves with 5k evaluation callbacks and linear trend overlays. DQN and PPO keep positive trends; A2C slightly drifts down.

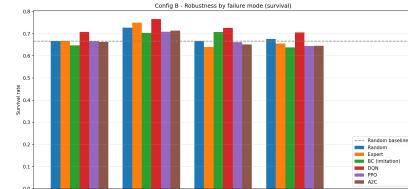


Figure 4: Survival by failure mode under the final 1M policies.

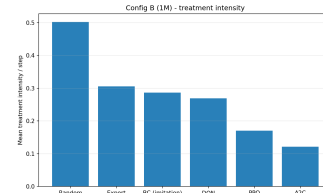


Figure 5: Treatment intensity: A2C is the most parsimonious; DQN trades moderate intensity for the highest survival.

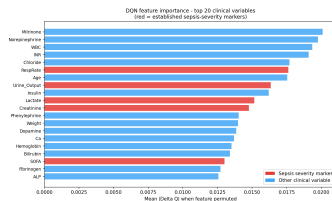


Figure 6: DQN feature importance; sepsis markers rank among the strongest signals.

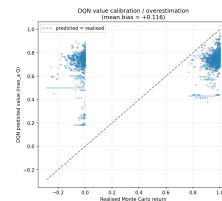


Figure 7: DQN value calibration; the positive bias motivates Double DQN.

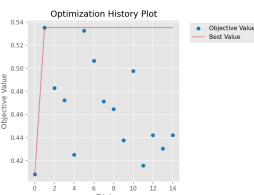


Figure 8: Optuna optimisation history for the DQN proxy search.

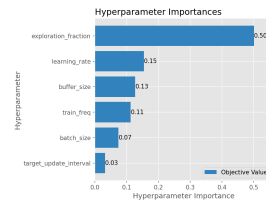


Figure 9: Optuna importance; learning rate and exploration fraction dominate.

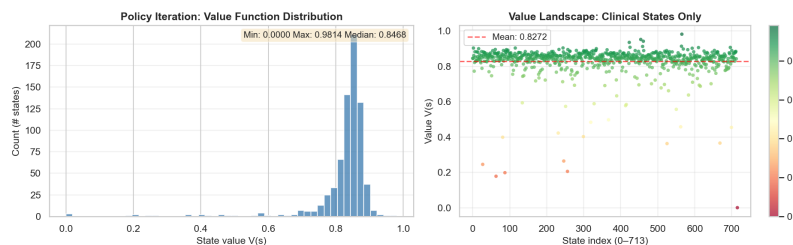


Figure 10: Configuration A: optimal value function $V^*(s)$ computed by Policy Iteration. Left: distribution of V^* across all 716 states. Right: per-state scatter coloured by value magnitude. This visualisation is unique to the tabular setting; the continuous Configuration B has no equivalent because the state space is infinite. The bimodal distribution separates low-value states (high SOFA, acute risk) from high-value states (stable patients close to discharge). The positive mean $\bar{V} \approx 0.7$ mirrors the $\sim 78\%$ survival ceiling.

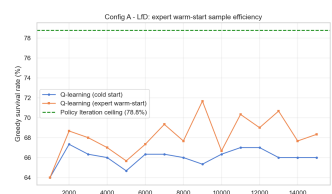


Figure 11: Expert Q warm-start (Configuration A): the warm-started agent escapes the sparse-reward plateau faster and finishes at higher survival than the cold start. PI ceiling shown as reference.

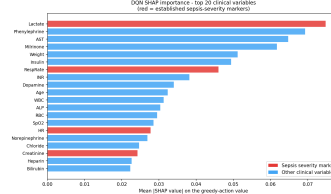


Figure 12: SHAP feature importance for the DQN. Established sepsis markers (lactate, urine output, respiratory rate, creatinine) rank among the top contributors, lending clinical plausibility to the policy.